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# Predicting Academic Performance of Student Using Classification Techniques

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***Abstract*—Data Mining methods are applied on educational data with the intent of enhancing teaching methods, improving quality of teaching, identifying weak students, identify factors that influence Student's academic performance. This utilization of data mining methods to elevate quality of education, identifying students who need improvement is termed as educational data mining. EDM has become a major research interest for many researchers. The primary function of educational data mining is prediction of student's academic performance. [1] Predicting student's academic performance helps in identifying a number of things like students who are likely to drop out, students who are weak and needs improvement, students who are good in academics but lately deteriorated. The intent of this paper is to determine factors that can influence a student's academic performance.**

***Keywords—Educational Data Mining, Performance Prediction, Classification, Decision Trees, WEKA.***

## I. Introduction

Education system all over the world has changed rapidly since vast research in the field of EDM and learning analytics. Use of DM methods, machine learning methods and different statistical techniques in education is EDM. EDM uses the above mentioned techniques to analyze data from educational institutions to uncover different patterns of student's behaviors and predict performance of students. The main goal of this research field is to help student improve their skills, to find out what hinders student from achieving success and how to improve it. This is done by various data mining techniques, machine learning algorithms and statistical techniques. Learning analytics is similar to Educational data mining, is a collection and analysis of usage data associated with student learning. [2]. Initially EDM and LA where major topics in education but slowly predicting student performance started becoming more popular as the primary objective of the topic is to analyze and predict student performance so that the student can perform better. This paper presents different classification methods by which performance of students can be predicted. The paper used dataset from UCI machine learning repository. The data attributes include student's marks, demographic, social and school related features. This dataset was previously used where different DM techniques such as Random Forest, Neural Networks, SVMs and regression techniques were used to predict secondary school student performance. They concluded that it is possible to predict final grades if previous grades are known. [3] The paper is further divided into five sections. Various related literature review is presented in the second section. Third section describes the implementation using WEKA tool, the methodology. Fourth section explains all the outcomes and findings. Conclusion is given in the fifth and last section of the paper.

## II. Literature Review

There are many papers published in the application areas of EDM & LA. Many researches have been done in Predicting academic performance of a student. In [4], a research was done to predict drop out students. The results proved that Decision Tress gives good accuracy. In [5], ANN technique was applied to predict academic performance of students. In [6], three classical DM methods namely Decision Tree, Naïve Bayes and Neural Networks were implemented and used to predict student academic achievement. Naïve Bayes produced better results. In [7], Kabakcheiva used CRISP-DM model to predict student academic performance. Decision Tree (J48) classifier, two Bayesian classifiers such as Naïve Bayes and BayesNet, Nearest Neighbor algorithm (IBk) and two rule learners (OneR and JRip) were applied. The results produced confirmed that Decision Tree (J48) has better accuracy than the rule learner (JRip) and IBk. Bayesian Classifiers had poor accuracy. In [8], regression analysis was applied to predict student's marks in a distance learning system. In [9], Genetic Programming was used with DM techniques to predict student failure. CRISP-DM is more popular with researchers as other DM process models such as SEMMA which is developed by the largest producers of analytics software, SAS Institute. [10]. It concentrates only on modeling tasks forgetting the business angles unlike CRISP-DM which has "Business Understanding Phase". SEMMA intends to provide services only to clients of SAS Enterprise Miner. [11]. In [12], C. Romero instead of using traditional classification algorithms to predict classes pass or fail, introduced and applied classification via clustering. In [13], Jia et al predicted student's retention by combining SVM and neural network to improve classification accuracy.

## III. Methodology

In this paper, we have used WEKA (Waikato Environment for Knowledge Analysis). It is a prominent data mining tool

written in Java, developed at the University of Waikato, New Zealand. It is non-propriety and freely available. [14]. We have used this tool for pre-processing, classification and visualization of the data.

TABLE I. NAMES AND DESCRIPTION OF ATTRIBUTES

| Attribute | Description | Value |
|---|---|---|
| SC | School | {GP,MS} |
| G | Gender | {F,M} |
| A | Age | 15-22 |
| AD | Type of Address | {U,R} |
| FS | Size of Family | {LE3,GT3} |
| PS | Parent's Cohabitation Status | {T,A} |
| ME | Educational Qualification of Mother | {None, Basic, Middle School, Secondary School, Graduate} |
| FE | Educational Qualification of Father | {None, Basic, Middle School, Secondary School, Graduate} |
| MJ | Employment type of Mother | {Educator, Healthcare, Civil Service, Home, Other} |
| FJ | Employment type of Father | {Educator, Healthcare, Civil Service, Home, Other} |
| RSN | Reason for opting a certain school | {distance, reputation, course preference, Other} |
| GD | Guardian of Student | {M, F, O} |
| TT | Time taken to travel to school | {<15min, 15-30 min, 30min-1hr, >1hr} |
| ST | Weekly Study time | {<2hrs, 2-5hrs, 5-10hrs, >10hrs} |
| FL | No of previous failures | n if 1<=n<3, else 4 |
| C | Coaching | {Y, N} |
| FES | Educational support given by family | {Y, N} |
| EPC | extra paid classes | {Y, N} |
| ECA | extra-curricular activities | {Y, N} |
| NUS | studied nursery | {Y, N} |
| HI | pursue higher education | {Y, N} |
| NET | Access to internet | {Y, N} |
| RS | Relationship Status i.e. In a relationship or not | {Y, N} |
| FR | Family relationship | {Very Bad, Bad, Medium, Good, Very Good} |
| FT | free time after school | {Very Low, Low, Medium, High, Very High} |
| HO | Hanging out with friends | {Very Low, Low, Medium, High, Very High} |
| DA | workday alcohol consumption | {Very Low, Low, Medium, High, Very High} |
| WA | weekly alcohol consumption | {Very Low, Low, Medium, High, Very High} |
| HL | current health status | {Very Bad, Bad, Medium, Good, Very Good} |
| AB | Absences in school | 0-93 |
| G1 | First year grade | 0-20 |
| G2 | Second year grade | 0-20 |
| Result(G3) | Final grade | {A,B,C,D,F} |

## IV. RESULTS AND DISCUSSIONS

In this segment, we will converse the outcomes and findings found from the output of Naïve Bayes classifier, J48 Decision Tree and MLP. Further we will analyze the dataset and compare attributes with final grades to see which attributes are strongest predictors. Performance of any algorithm is evaluated on the basis of precision and recall. Classification matrix of J48, Naïve Bayes and MLP is shown in Table II, Table III and Table IV. Also TP rate, FP rate, Precision and Recall of each classifier is shown in Table V, Table VI and Table VII. Table VIII shows the comparison of three classifiers on the basis of time taken to run and correctly classified instances.

TABLE II. CONFUSION MATRIX OF J48

| G3 | | Prediction | | | | |
|---|---|---|---|---|---|---|
| | | Fail | D | B | A | C |
| Actual | Fail | 109 | 21 | 0 | 0 | 0 |
| | D | 16 | 70 | 0 | 0 | 17 |
| | B | 0 | 0 | 46 | 3 | 11 |
| | A | 0 | 0 | 11 | 29 | 0 |
| | C | 0 | 12 | 12 | 0 | 38 |

TABLE III. CONFUSION MATRIX OF NAÏVE BAYES

| G3 | | Prediction | | | | |
|---|---|---|---|---|---|---|
| | | Fail | D | B | A | C |
| Actual | Fail | 93 | 37 | 0 | 0 | 0 |
| | D | 11 | 69 | 3 | 0 | 20 |
| | B | 0 | 0 | 48 | 6 | 6 |
| | A | 0 | 0 | 11 | 28 | 1 |
| | C | 0 | 16 | 13 | 0 | 33 |

TABLE IV. CONFUSION MATRIX OF MLP

| G3 | | Prediction | | | | |
|---|---|---|---|---|---|---|
| | | Fail | D | B | A | C |
| Actual | Fail | 93 | 33 | 0 | 0 | 4 |
| | D | 34 | 44 | 7 | 0 | 18 |
| | B | 1 | 6 | 26 | 13 | 14 |
| | A | 0 | 2 | 12 | 23 | 3 |
| | C | 4 | 28 | 10 | 4 | 16 |

TABLE V. PERFORMANCE OF NAÏVE BAYES

| CLASSIFIER | TP RATE | FP RATE | PRECISION | RECALL | CLASS |
|---|---|---|---|---|---|
| NB | 0.700 | 0.017 | 0.824 | 0.700 | A(16-20) |
| | 0.800 | 0.081 | 0.640 | 0.800 | B(14-15) |
| NB | 0.532 | 0.081 | 0.550 | 0.532 | C(12-13) |
| | 0.670 | 0.182 | 0.566 | 0.670 | D(10-11) |
| | 0.715 | 0.042 | 0.894 | 0.715 | FAIL |

TABLE VI. PERFORMANCE OF MLP

| CLASSIFIER | TP RATE | FP RATE | PRECISION | RECALL | CLASS |
|---|---|---|---|---|---|
| MLP | 0.575 | 0.048 | 0.575 | 0.575 | A(16-20) |
| | 0.433 | 0.087 | 0.473 | 0.433 | B(14-15) |
| | 0.258 | 0.117 | 0.291 | 0.258 | C(12-13) |
| | 0.427 | 0.236 | 0.389 | 0.427 | D(10-11) |
| | 0.715 | 0.147 | 0.705 | 0.715 | FAIL |

TABLE VII. PERFORMANCE OF J48

| CLASSIFIER | TP RATE | FP RATE | PRECISION | RECALL | CLASS |
|---|---|---|---|---|---|
| J48 | 0.725 | 0.008 | 0.906 | 0.725 | A(16-20) |
| | 0.767 | 0.069 | 0.667 | 0.767 | B(14-15) |
| | 0.613 | 0.084 | 0.576 | 0.613 | C(12-13) |
| | 0.680 | 0.113 | 0.680 | 0.680 | D(10-11) |
| | 0.838 | 0.060 | 0.872 | 0.838 | FAIL |

TABLE VIII. PERFORMANCE OF COMPARISON

| | NAÏVE BAYES | MLP | J48 |
|---|---|---|---|
| TIME TAKEN TO BUILD MODEL | 0.03 SECONDS | 29.06 SECONDS | 0.17 SECONDS |
| CORRECTLY CLASSIFIED INSTANCES | 68.6076 % | 51.1392 % | 73.9241 % |
| INCORRECTLY CLASSIFIED INSTANCES | 31.3924 % | 48.8608 % | 26.0759 % |

Clearly, we can see that G1 and G2 are the strongest predictors. But we already knew it as student achievement is highly affected by previous performances. [15]. Let us see other variables which might influence final grades by doing explanatory data analysis. Drop out Students have been identified. There were no students with score "0" in first period i.e. G1. Number of students who scored only zero in second period i.e. G2 were 13 but these same students also scored "0" in final period meaning all these students are drop-outs. One interesting pattern among drop out students is that all the students studied in the same school i.e. GP. Almost all students fall in 15-16 age categories. We removed G1, G2 and failed class variables which are strongest predictors as we are trying to figure out what other factors other than previous academic performances that can influence final grades. Fitting Random Forest Model, we find out in variable importance plot as shown in Fig 1 the top factors influencing final grades are students who wants to do higher education, student's study time, weekend and workday alcohol consumption, mother's education and romantic relation of Student. Variables compared with final grades are given in Table VIII.

TABLE IX. VARIABLES COMPARED WITH FINAL GRADES

| S. No. | Variables compared with Final Grade |
|---|---|
| 1 | Alcohol Consumption(Workday and Weekend) |
| 2 | Health |
| 3 | Romantic Relationship |
| 4 | Parent's Education |

*Alcohol Consumption*

When alcohol consumption given in dataset is compared with final grade, we found out that alcohol consumption in week days does affect academic performance. Although weekend alcohol consumption does not affect grade as much as weekday alcohol consumption, we can see this in Figure 2 and Figure 3. We also compared both "WA" and "DA" with absences and found out that more drinking leads to more absences. So we can clearly say that alcohol consumption does affect academic performance.

*Health*

One interesting finding is that student with poor health status scored more than the students who had a health score of "very good". So we can't conclude that healthier students tend to score high.

*Romantic Relationship*

When we compared romantic relationships with academic performance, we found out mean final grade of those students who are not in a romantic relationship is greater than those who are in a romantic relationship. Students who are not in relationship scored higher grades more frequently than those students who are in relationship

*Parents Education*

We wanted to see how Parent's Education affects the final grade. When both Mother and Father's Education were compared with final grades and we found out that both parent's education level is important factor for scoring good grades. As the education level of mother increases, the desire to pursue higher education increases. We observe that as the education level of mother and father increases, the probability of having zero number of failures decreases.

When parent education level is compared with internet access and study time, some good observations were recorded. As the education level of mother increases, the probability of student having internet access increases. The mean study time increases with the increase in mother's education level. For father's case there is no such pattern. Also, we found out that when the education level of the mother increases, the percentage of students who consume high level of alcohol during weekends decreases.

Thus, we understand how much important factor is mother education for student's scoring good grades. So we can conclude that alcohol consumption (weekend and workday both), parent's education are strong predictors.

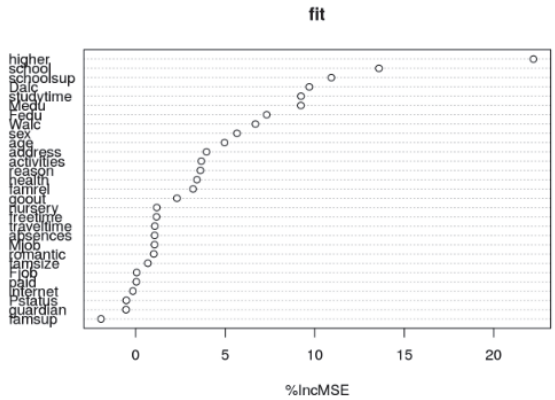


Fig. 1. Variable Importance plot

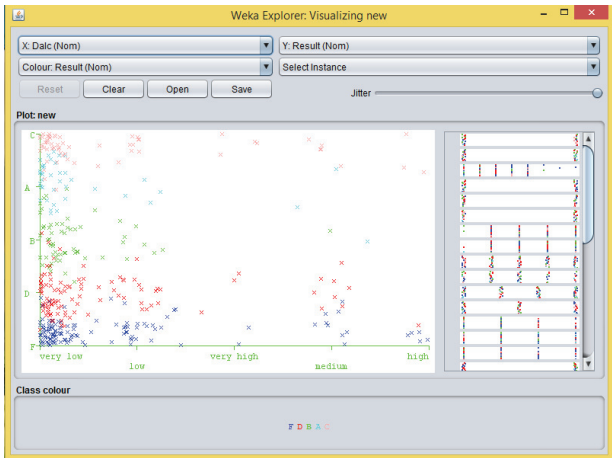


Fig. 2. Weekday Alcohol Consumption

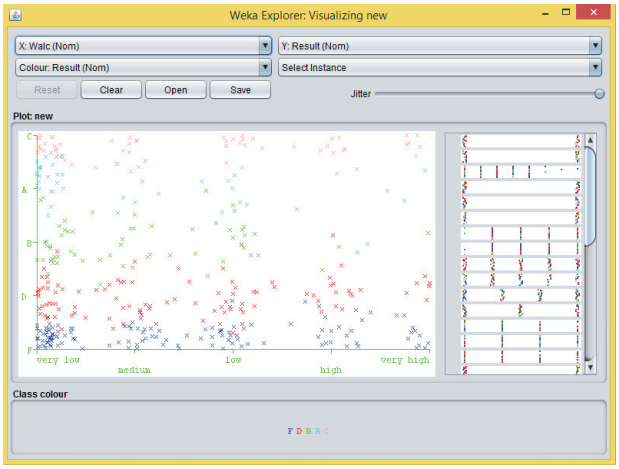


Fig. 3. Weekend Alcohol Consumption

## V. Conclusion

Predicting student's academic performance is of great concern to the education institutes. It helps us to identify the abilities of students, their interests and weaknesses. Student performance can be influenced by different types of attributes. This can be social, demographic and related to school. Here we used Naïve Bayes, J48 decision tree and MLP for classification. Naïve Bayes had 68.60 % accuracy, J48 had 73.92% accuracy and MLP had 51.13% accuracy. Previous grades are highly linearly correlated to final grades. Without previous grades it is very difficult to predict final grades but we can analyze the data and find which other attributes plays an important role in predicting final grades. The top factors influencing final grades are weekend and workday alcohol consumption, Parent's education. Also romantic relationship slightly influences academic performance. Education level of mother is another important factor in determining student's good scores.

## References


[1] C. Romero, S. Ventura, "Educational Data Mining: A Review of the State of the Art," IEEE Transactions on Systems, Man, and Cybernetics—Part c: Applications and Reviews, vol. 40, no. 6, 2010, pp. 601-618.

[2] G. Siemens, Learning and Academic Analytics, 5 August 2011, http://www.learninganalytics.net/?p=131

[3] P. Cortez and A. Silva. "Using Data Mining to Predict Secondary School Student Performance", In A. Brito and J. Teixeira Eds., Proceedings of 5th FUture BUsiness TEChnology Conference (FUBUTEC 2008) pp. 5-12, Porto, Portugal, April, 2008, EUROSIS, ISBN 978-9077381-39-7.

[4] G.W.Dekker,M. Pechenizkiy and J. M. Vleeshouwers, "Predicting Students Drop Out: A Case Study", 2nd International Conference On Educational Data Mining Proceedings, Cordoba, Spain, July 1-3, 2009

[5] M. F. Musso, E. Kyndt, E. C. Cascallar and F. Dochy, "Predicting general academic performance and identifying the differential contribution of participating variables using artificial neural networks", Frontline Learning Research, 1(1):42–71, 2013.

[6] E. Osmanbegović , M. Suljić, "DATA MINING APPROACH FOR PREDICTING STUDENT PERFORMANCE", Journal of Economics and Business, Vol. X, Issue 1, May 2012

[7] D. Kabakchieva, "Predicting Student Performance by Using Data Mining Methods for Classification", Cybernetics and Information Technologies- The Journal of Institute of Information and Communication Technologies of Bulgarian Academy of Sciences, DOI: 10.2478/cait-2013-0006

[8] S. B. Kotsiantis, "Use of machine learning techniques for educational proposes: a decision support system for forecasting students grades",Artificial Intelligence Review, 37(4):331–344, 2012.

[9] C.M. Vera, A. Cano, C. Romero, S. Ventura, "Predicting student failure at school using genetic programming and different data mining approaches with high dimensional and imbalanced data", DOI 10.1007/s10489-012-0374-8

[10] A. Azevedo, and M. F. Santos, "KDD, SEMMA and CRISP-DM: a parallel overview" In Proceedings of the IADIS European Conference on Data Mining 2008, pp 182-185. Archived January 9, 2013, at the Wayback Machine.

[11] S. S. Rohanizadeh and M. B. Moghadam, "A Proposed Data Mining Methodology and its Application to Industrial Procedures", Journal of Industrial Engineering 4 (2009) pp 37-50.

[12] C. Romero, M. López, J. M. Luna, S. Ventura, "Predicting students' final performance from participation in on-line discussion forums", ELSEVIER, Computers & Education 68 (2013) 458–472, 2013

[13] J.W. Jia and M. Mareboyana,"Machine Learning Algorithms and Predictive Models for Undergraduate Student Retention", In Proceedings of the World Congress on Engineering and Computer Science, volume 1, 2013.

[14] M. Hall, E. Frank, G. Holmes, B. Pfahringer, P. Reutmann, & I. Witten (2009), "The WEKA Data Mining Software: An Update", ACM SIGKDD Explorations Newsletter, 11(1), 10-18. Retrieved from http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.148.3671

[15] S. Kotsiantis, C. Pierrakeas and P. Pintelas (2004), "Predicting Students' Performance in Distance Learning Using Machine Learning Techniques", Applied Artificial Intelligence (AAI), 18, no. 5, 411–426.